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Course : Data Analytics with SQL

Exercise 1 : SQL Foundations

1. Selecting and Sorting.

1. SELECT *
FROM Table_employees.

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Table_employees.

ID	first name	last name	Department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Capetown
3	Cathy	Zulu	finance	65000	2018-07-18	Durban
4	David	Morcoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. SELECT Distinct Department
FROM Table_employees ;

↓

Table_employees

Department
IT
HR
finance
Marketing

3. SELECT first_name,
last_name
FROM Table_employees
ORDER BY Salary_Desc;

↓

Table_employees

First_name	last_name
Brian	Lee
David	Moroena
Cathy	Zulu
Alice	Green
Eva	Naidoo

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4. SELECT *
FROM Table_employees
~~WHERE Top 3 highest paid~~
ORDER BY Salary_ASC
LIMIT 3

↓

Table_employees

ID	First_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

2. Filtering with WHERE

5. SELECT Distinct
FROM table_employees
WHERE Department = IT

ID	First_name	last_name	department	Salary
1	Alice	Green	IT	70 000
5	Eva	Naidoo	IT	72000

6. SELECT ID,
First name,
last name,
Department,
Salary
FROM table_employees
WHERE salary > 60000

ID	First name	last_name	Department	Salary
1	Alice	Green	IT	70 000
3	Cathy	Zulu	finance	65 000
5	Eva	Naidoo	IT	72 000

7. SELECT ID,
First-name,
last-name,
Department
FROM table_employees
WHERE Employees in HR OR Marketing

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ID	First_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT ID
first_name
last_name
Department
FROM table_employees
WHERE Department Not in IT

ID	first name	last name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	finance
4	David	Mokoena	Marketing

9. SELECT ID,
first name,
last name,
Department
FROM table- employees
WHERE Department IN (IT, HR, OR finance);

↓

ID	First name	last name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT ID,
first name,
last name,
Department,
Salary,
City
FROM table- employees
WHERE Department IN IT with Salary > 65000 AND City = Johannesburg

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ID	first name	last name	Department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg